Metaplectic Jacquet–Langlands Transfer, Companion Paper (Rank Two, Wild Place)
Revision Log: Draft 1 to Draft 2

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Scope

Draft 2 responds to ChatGPT's referee report on Draft 1 (recommendation: major revision). The report accepted the finite Fourier calculations after independent reconstruction, but did not accept Target A as proved, for two reasons concerning Proposition `prop:H1`: the written proof of case (i) contained an incorrect assertion ($\beta(X, X') = 0$ whenever $v(t) > 0$), exhibited by the explicit pair $X = 1 + \varpi^3 j$, $X' = 1 + \varpi^3 + \varpi^4 j$ with $\beta = 2$; and cases (iv)–(v) rested on a stratified sample rather than a proof. The referee asked that (H1) be repaired without new hypotheses, that the proof of Target A then be reexamined for remaining dependencies, that the finite Fourier work be preserved, and that the Python modules be supplied.

The repair of (H1)

The referee's example is correct: in the explicit model the pair above has $\beta = 2$, $\kappa(X) = 0$, $\kappa(X') = 1$, $\kappa(XX') = 0$, so the identity holds but the Draft 1 argument for it was wrong. The error was the claim that $1 + \varepsilon$ in $r = (1 + t)(1 + \varepsilon)$ lies in $U^{(4)}$; it lies only in $U^{(3)}$, and $(U^{(3)}, \varpi)_{E_0} \neq 1$.

Draft 2 replaces the case split entirely. Two changes make a complete hand proof possible.

1. **A cleaner closed form for the star section.** For $X = x + yj \in K^6$ write $X = zv^3$ with $z \in U_F^{(1)}$, $v \in K^2$. Taking reduced norms gives $z \equiv \mathrm{Nrd}(X)^{-1}$ modulo $F^{\times 3}$, so the star lift is $\ell(X) = (X, \kappa(X))$ with

$$\kappa(X) = (y, \mathrm{Nrd}\, X)_{E_0} \quad (y \neq 0), \qquad \kappa(X) = 0 \quad (y = 0),$$

   (Corollary `cor:starK6`). This is equivalent to the Draft 1 digit formula $c(X)\, v(y)$ but needs no digits, and it is what makes the identity below transparent.

2. **The right factorization.** In the main case ($y, y' \neq 0$, $XX' \notin E_0$), with $\tau = \sigma(y'/y)$ and $R = x' + \sigma(x)\tau$, the identity (H1) is equivalent to

$$(R, \tau)_{E_0} = v(\tau)\lambda(x') - v(R)(\lambda(x') - \lambda(x)), \qquad \lambda(x) := (x, \varpi)_{E_0}.$$

   Factoring $R = x'(1 + \tau_2)$ with $\tau_2 = \sigma(x)\tau/x'$ and writing $\tau = \tau_2 \cdot x'/\sigma(x)$, the left side splits into three symbols: (x', τ), which is $v(\tau)\lambda(x')$ because $x' \in U^{(3)}$ kills μ_8 and $U^{(1)}$; $(1 + \tau_2, \tau_2) = 0$ by Steinberg; and $(1+\tau_2, x'/\sigma x)$, which is $v(R)$ times $(\varpi, x') - (\varpi, \sigma x)$ by the same depth bound and

Galois invariance. No case distinction on $v(\tau)$ or on $v(1+\tau)$ occurs; the former boundary cases (iv) are not special. The degenerate cases ($y = 0$, $y' = 0$, both, or $XX' \in E_0$) are each a few lines of the same kind. The only inputs are Kubota's cocycle formula, bimultiplicativity, antisymmetry, Steinberg, Galois invariance, triviality on cubes, and the depth bound $(U^{(i)}, U^{(j)})_{E_0} = 1$ for $i + j \geq 4$ (Fesenko–Vostokov; Serre). No new hypothesis is introduced, and the explicit values of the symbols are not used in this proof.

Proposition `prop:H1` is now labeled [P]. A new script (`step12_H1_proof_check.py`) reproduces the referee's example, checks the new closed form against the digit form on 3000 elements, checks the symbol facts used, and checks the reduced identity and (H1) on 12,000 structured instances covering every case of the proof ($\beta \neq 0$ in about 44%); this is corroboration only.

Reexamination of Target A: dependency ledger

A new Remark `rem:dependencies` in Section 11 itemizes everything the main theorem uses:

(D1) [E] Part I (Definition 2.2, Propositions 2.3, 3.1, 3.5(a), Lemma 3.4, Theorem 3.9, Remark 8.1) and Part III (normalization of L_D, $b_D = 3$, $c_D = 27$, $R = 1$ on 2-power elements).

(D2) [E] Kubota's cocycle on $\mathrm{GL}_2(E_0)$ and the standard properties of the cubic Hilbert symbols.

(D3) [P^c] the explicit symbol tables (exact linear algebra via Moore's theorem), used only through $(1 + \varpi^3, \varpi) \neq 1$ in the exclusion of $e = 0$.

(D4) [P^c] four exhaustive finite enumerations: $\bar{G}_3$ from the 648 depth-one classes; the cube map on Q/Q_7; the actions on V; the design search with the Heisenberg extension and Fourier identities. The referee reconstructed the design data independently.

(D5) [P] everything else, including (H1).

No sampled computation is load bearing. The Lazard-splitting route of Draft 1 is retained as a remark (second route), no longer needed.

Status changes

Item	Draft 1	Draft 2
Lemma `lem:model` (the model is the cover)	[P^c]	[E]/[P] (cocycle cited, not sampled)
Corollary `cor:starK6`	[P]	[P] (new closed form)
Proposition `prop:H1`	[P^c], case (i) wrong	[P]
Target A	[P^c] with one sampled step	[P^c], all computations exhaustive
Level-zero rows and norm twists	[P^c]	[P^c]
Target B positive level; Target C	[O]/[C]	[O]/[C]

The label [P^c] on Target A now means exactly (D3)–(D4): finite enumerations over explicitly described sets, each with a supplied exact script. The definition of the label in Section 1 was tightened accordingly ("exhaustive" is now part of it).

Items from the referee report not yet addressed

The report's table marks Sections 7 and 8 "promising; some arguments require fuller justification" and "minor gaps to repair" without specifics in the portion received. The measure-theoretic step in Lemma `lem:cube`(f) (cubing is a local homeomorphism near regular elements) is the place in Section 7 most likely meant; it is corroborated by script 2 in Q/Q_7 but could be replaced by a direct Hensel argument. Section 8 was not changed in Draft 2. Both are candidates for Draft 3 once the referee's specifics are available.

Editorial changes

Abstract and Section 1 updated; status table splits the Section 10 row into the model lemma ([E]/[P]) and the proofs ([P]); Remark `rem:H1-status` removed and replaced by Remark `rem:H1-checks`; Remark `rem:dependencies` added; Appendix B lists script 12; bibliography adds Kubota, Fesenko–Vostokov, Serre, Symonds–Weigel. The acknowledgments record that the Draft 1 error was found by ChatGPT's report. Sections 2–9 and the finite Fourier material are unchanged, as the referee requested. American English spelling throughout.

Deliverables

`mjl_rank2_wild_draft2.tex/.pdf` (27 pages), this log, and `mjl_rank2_wild_scripts.tar.gz` containing `quat.py`, `cover.py` (exact $\mathbb{Q}(\zeta_{12})$ arithmetic; the cubic Hilbert symbols are solved in `step7_symbols.py` from the Steinberg relations and stored in `symbols.pkl`), and `step1`–`step12`. To verify the cocycle and symbol implementation independently, run `step7_symbols.py` first (regenerates `symbols.pkl`), then `step12_H1_proof_check.py`.